

Comment on Papers using Machine Learning for Significant Wave Height Time Series Prediction: Complex Models Do Not Outperform Auto-regression

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Highlight:

- Five Machine Learning (ML) models compared for wave height time series prediction
- Complex ML models do not outperform simple AR in wave height time series prediction
- Comment to related papers: signal decomposition in test set series is WRONG

Key words:

Significant wave height; Time series prediction; Machine learning; Deep learning

Abstract

Significant Wave Height (SWH) is crucial in many aspect of ocean engineering. The accurate prediction of SWH has therefore been of immense practical value. Recently, Artificial Intelligence (AI) time series prediction methods have been widely used for single-point short-term SWH time-series forecasting, resulting in many AI-based models claiming to achieve good results. However, the extent to which these complex AI models can outperform traditional methods has largely been overlooked. This study compared five different models - AutoRegressive (AR), eXtreme Gradient Boosting (XGB), Artificial Neural Network (ANN), Long Short-Term Memory (LSTM), and WaveNet - for their performance on SWH time series prediction at 16 buoy locations. Surprisingly, the results suggest that the differences of performance among different models are negligible, indicating that all these AI models have only “learned” the linear auto-regression from the data. Additionally, we noticed that many recent studies used signal decomposition method for such time series prediction, and most of them decomposed the test sets, which is WRONG.

1 Introduction

Significant Wave Height (SWH) is a crucial wave parameter for the safety of human activities in the ocean, such as ocean engineering, marine navigation, and exploration and exploitation of marine resources. The accurate prediction of SWH has always been an essential issue (e.g., Deo and Naidu 1998; The WISE Group 2007; Fan et al. 2020).

Numerical Wave Models (NWMs) based on the wave action balance differential equation are the most widely used method for wave prediction (e.g., Tolman 1991; Booij et al. 1999). After years of development, NWMs can predict wave parameters well if high-quality prediction of wind and current forcing is available (e.g., The WISE Group, 2007; Liu et al., 2019; Alday et al., 2021), and have been widely applied to operational wave forecasts. However, due to their complex calculation of the propagation and evolution of the full directional wave spectra, they cost a large amount of computing resources and time. These make NWMs sometimes unsuitable for time-sensitive applications that need quick predictions, especially considering that high-quality wind and current predictions may also be unavailable in some cases.

In recent years, with the rapid development of machine learning and deep learning technologies, studies started to use artificial intelligence (AI) as alternative methods for wave predictions (e.g., Liu et al. 2024, Wang et al. 2023, Song and Jiang 2023). One direction of the AI-based wave prediction is using AI to forecast short-term (0~72h) SWH time series at fixed oceanic locations using historical SWH time series as input. Many machine learning models such as Artificial Neural Network (ANN, e.g., Deo and Naidu 1998), locally weighted regression (e.g., Zamani et al. 2008), and support vector machine (e.g., Mahjoobi and Mosabbebi 2009), have been tried. With the concept of “deep learning” becoming popular (Lecun et al. 2015), many recent studies begin to use much complex frameworks of deep neural networks such as Long Short-Term Memory (LSTM) (e.g., Altunkaynak 2015, Fan et al. 2020, Ni and Ma 2020, Huang and Dong 2021, Gao et al. 2021, Zhou et al. 2021, Çelik 2022, Feng et al. 2022, Song et al. 2023, Chen et al. 2023, Altunkaynak et al. 2023, Çelik and Altunkaynak 2023, to name but a few). These works are more or less in the same way, except that different deep learning model architectures are selected (some of them decompose the time series before modelling, which is wrong as explained in Section 4).

These studies all claim that AI methods can automatically “learn” the patterns and modes of SWH series from the input historical data, and can well predict future SWHs.

This study can be regarded as a comment on these AI-based SWH time series prediction papers. Predicting SWH time series using historical SWH is a typical univariate time series prediction problem, and the essence of time series prediction is actually to explore the linear or nonlinear autocorrelations of the sequence, and predict future data based on them. A very simple but widely used method for such time series prediction problems is the autoregressive (AR) model, which can capture the linear autocorrelation of the sequence. While sophisticated models like deep learning are often utilized, simpler models such as the autoregressive (AR) model can perform equally well, especially when dealing with a limited number of input variables. However, we realized that almost no studies of AI-based SWH time series prediction (none of the aforementioned ones) have compared their results with the AR model with the same inputs (probably simply because the AR is too simple). Therefore, this study aims to compare the performance of different time series prediction models, including AR, eXtreme Gradient Boosting (XGB), ANN, LSTM, and Wavenet, to predict SWH time series at different locations. Based on 16 coastal or open-ocean buoys with different wave climate characteristics in their locations, each model was trained and tested using input lengths of 6h and 24h, and prediction lengths of 1h, 3h, 6h, 12h, 18h, 24h, 36h, 48h, and 72h, respectively. We hoped this comparison can provide reference and support for practical marine applications. The results, however, are surprising to some extent: the performance of AI models is not significantly better than that of the simplest AR model.

2 Data and Models

2.1 Data

Hourly SWH time series of 16 buoys from the National Data Buoy Center (NDBC, <https://www.ndbc.noaa.gov/>) over the period 2016-2017 were used in this study (Figure 1). The locations of these buoys include both nearshore and offshore regions, both wind-sea-dominated and swell-dominated regions, and contain water depths ranging from 10 to 4000 m. The annual mean SWHs of these locations range from 0.5 to 2.5 m, with corresponding SWH standard deviations ranging from 0.4

to 1.4 m and coefficients of variation (standard deviation/mean, reflecting the relative degree of variation in SWH) ranging from 0.2 to 0.65. Therefore, the locations of these buoys can well represent regions with many different wave climates.

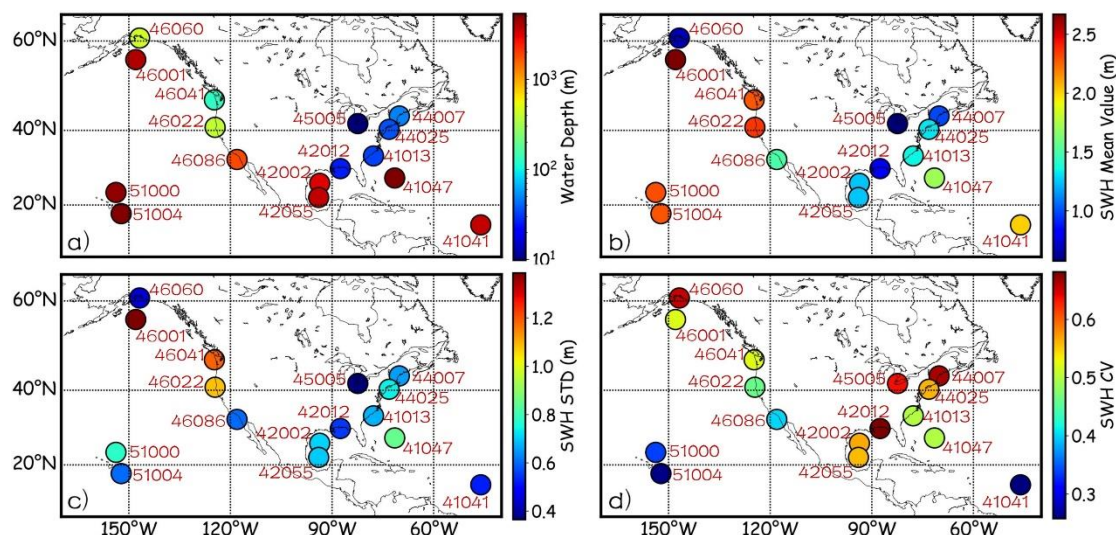


Figure 1. The distribution of the NDBC buoys used in this study. The color indicates the (a) water depth, (b) annual SWH mean, (c) SWH standard deviation, and (d) SWH coefficient of variation at each buoy location.

2.2 Models

Five types of models were used in this study, including AR, XGB, ANN, LSTM, and Wavenet. Among them, the AR is the simplest model for time series prediction, and the XGB and ANN are the simplest tree-structure regression model and the simplest deep learning regression model, respectively, and the LSTM model are widely used in previous studies of SWH time series forecast (e.g., Fan et al. 2020, Ni and Ma 2020, Huang and Dong 2021, Gao et al. 2021, Zhou et al. 2021, Feng et al. 2022, Song et al. 2023, Chen et al. 2023). For readability and to save space, we do not expand the rationales of these models in the main text. Brief introductions of these models, as well as how they are used in this study, can be found in the Supplemental Materiel. One can also easily find their introductions in textbooks or online lectures on machine learning and deep learning (e.g., Goodfellow et al. 2016), and they can all be easily realized using Python packages such as *torch* (<https://pytorch.org/>, details of Packages are available on the websites) (for ANN, LSTM, and Wavenet), and *xgboost* (for XGB) (<https://xgboost.readthedocs.io/>). The Python codes to realize these models using the data set in this study are also given in the Supplemental Materiel.

For each buoy, the data in 2016 and 2017 were used as the training test and validation test, respectively. We attempted to predict SWH at 1 h, 3 h, 6 h, 12 h, 18 h, 24 h, 36 h, 48 h, and 72 h after forecast start time, using input sequences of 6 h and 24 h before forecast start time. The reason for selecting these input and output lengths is simply that they are also widely used in previous studies. For each combination of buoy, input length m , and forecast length n , the five models were trained separately using the corresponding data. This means that $16 \text{ (buoys)} \times 9 \text{ (forecast lengths)} \times 2 \text{ (input lengths)} \times 5 \text{ (model types)} = 1440$ models were trained. Or to say, for each buoy and each model type (e.g., AR, ANN, LSTM), the input can be 6 h or 24 h (two input lengths) SWH sequence before forecast start time. For each input length, nine separate models are trained for nine different target outputs (nine sets of forecast lengths). Figure 2a shows the schematic of the inputs and outputs for each of the 1440 models. The green time points are the inputs of the model with a length of m hours (input length), and the yellow time points are the outputs of the model which corresponds to the forecast lengths of n hours. Figures 2b and 2c show two examples for $m=24/n=6$ and $m=6/n=24$, respectively.

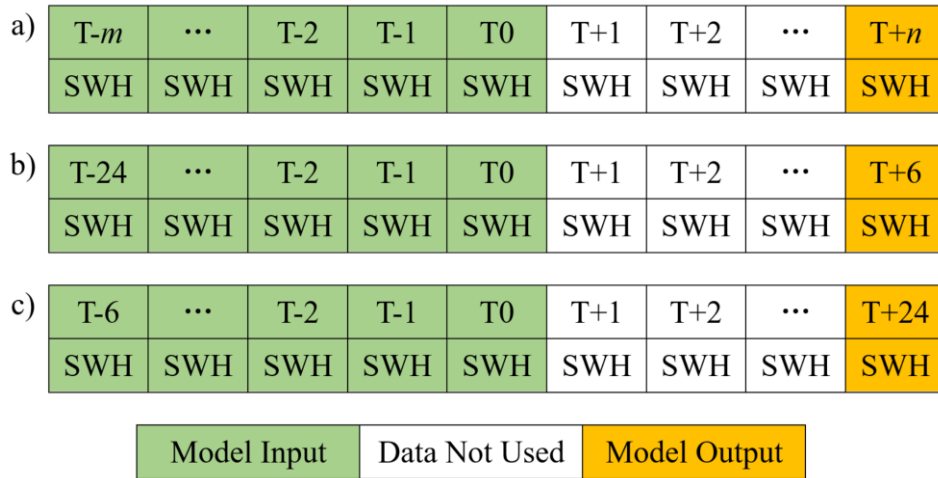


Figure 2. A schematic of the inputs and outputs of SWH prediction models used in this study. The inputs and outputs of the model are displayed in green and yellow, respectively. Subplot (a) shows a general case of an input length of m hours and forecast length of n hours. Subplots (b) and (c) show two examples for $m=24/n=6$ and $m=6/n=24$, respectively.

All models were trained to minimize the Root Mean Square Error (RMSE) between the predicted SWH and the buoy-observed SWH. Except for the AR model, all other model types involve the selection of various hyper-parameters, such as the number of hidden layers, and the numbers of neurons in different layers. We conducted some tests to select the structure of the networks, and for each type of model, the

hyper-parameters are the same for all buoys, forecast lengths, and input lengths. The reported results in this study reflect the best outcomes obtained through multiple rounds of experiments. Detailed model settings and hyper-parameters are given in the Supplemental Material along with the Python codes in which the model settings can also be found. We also found that the performance of the model is actually insensitive to these hyper-parameters, as similar results can be easily replicated using different settings.

All the deep learning models were trained using the Adam optimizer with a batch of 256. The early stopping and dynamic learning rate strategies were used to prevent over-fitting: The learning rate (initially set to 0.004) was decreased by 50% if the RMSE of the training set did not decrease for two epochs, and the training process stopped when the RMSE of the validation set did not decrease for six epochs and the resulting best-performed model was selected.

3 Results

The Correlation Coefficient (CC) and Root Mean Square Error (RMSE) of the test set were selected as the error metrics to evaluate the performance of the models:

$$CC = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2} \quad (2)$$

where x and y denote the model-predicted and buoy-observed SWH, respectively, and the bars over them denote their mean values. Here, we do not use the bias simply because all such data-driven statistical models are unbiased. The errors can also be described using more error metrics such as normalized RMSE, mean absolute error, and normalized mean absolute error. The result for the other error metrics are only displayed in the SM to save space since they do not affect any conclusions.

Here, we take buoy 51004 as an illustrative example. Tables 1 and 2 show the CC and RMSE values, respectively, of different types of models for predicting the SWH at the buoy's location for different input and forecast lengths. The results clearly show that the errors of the predicted SWH increase significantly as forecast length extends from 1 to 72 h: CCs decrease from over 0.95 to below 0.5, and RMSEs increase from below 0.15 m to over 0.5 m. Compared to the impact of the forecast length on the accuracy of the predicted SWH, the impact of input lengths on the results appears negligible. We also tried several different input lengths from 9 to 72 h, which gives the same conclusion. Similarly, given the forecast

length, the performance of the five types of models does not exhibit apparent differences. The difference in CCs (RMSEs) between the best model and the 6h-input AR model is less than 0.01 (0.01 m) for all combinations of forecast and input lengths.

Table 1. The CCs for different SWH-prediction models with different input and forecast lengths at the location of buoy 51004.

CC	Input Length: 6h					Input Length: 24h				
Forecast Time	AR	XGB	ANN	LSTM	Wavenet	AR	XGB	ANN	LSTM	Wavenet
1h	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97
3h	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96	0.96
6h	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93	0.93
12h	0.85	0.85	0.85	0.85	0.85	0.86	0.85	0.85	0.85	0.85
18h	0.77	0.77	0.77	0.78	0.78	0.77	0.78	0.77	0.78	0.78
24h	0.70	0.7	0.7	0.71	0.71	0.70	0.70	0.71	0.71	0.71
36h	0.58	0.58	0.57	0.58	0.58	0.59	0.57	0.57	0.57	0.59
48h	0.51	0.51	0.51	0.51	0.51	0.53	0.52	0.52	0.51	0.52
72h	0.45	0.45	0.44	0.46	0.43	0.46	0.43	0.46	0.44	0.45

Table 2. The same as Table 1, but for RMSE (Unit: m).

RMSE	Input Length: 6h					Input Length: 24h				
Forecast Time	AR	XGB	ANN	LSTM	Wavenet	AR	XGB	ANN	LSTM	Wavenet
1h	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
3h	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17
6h	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22
12h	0.31	0.31	0.31	0.31	0.31	0.31	0.31	0.32	0.31	0.31
18h	0.38	0.38	0.38	0.37	0.37	0.38	0.37	0.38	0.37	0.37
24h	0.43	0.43	0.43	0.42	0.42	0.43	0.43	0.42	0.42	0.42
36h	0.49	0.48	0.49	0.48	0.48	0.49	0.50	0.50	0.50	0.48
48h	0.52	0.52	0.52	0.52	0.52	0.51	0.51	0.52	0.51	0.51
72h	0.54	0.54	0.54	0.53	0.53	0.54	0.54	0.54	0.53	0.53

We performed similar calculations for the rest 15 buoys and obtained similar results that demonstrate the negligible impact of input length and model selection on prediction results. The variation of CC between the 6-h-input AR model and different models with different input lengths for the same forecast length is less than 0.03. To save space, we do not display all error metrics in tabular form here. Instead, Figure 3 shows the changes in CCs and RMSEs of different types of models with varying forecast lengths at the 16 selected buoy locations with the input length fixed to 24h. For the same model, the performance of the SWH forecast varies significantly at different locations, due to the auto-correlation characteristics of the sequences themselves. For example, the waves at buoy 51000 near Hawaii are mainly stable swells from the westerlies in both hemispheres, leading to a relatively smooth time series with a relatively long autocorrelation length, and a long autocorrelation length corresponds to better predictability over a short period. The waves at buoy 41047 are stable waves generated by the trade winds, which are also easier to

predict using historical data. The CC for the 12-h forecast of the SWH at these buoys can be larger than 0.8 using all models. In contrast, buoy 45005 in Lake Erie is almost entirely controlled by rapid-response local wind waves, and the corresponding short autocorrelation length makes its predictability of SWH extremely poor. The CC for the 6-h forecast of the SWH at 45005 is already less than 0.75, and is less than 0.5 for the 12-h forecast. Meanwhile, for any given location, the prediction error increases significantly with increasing forecast lengths but there is no discernible difference in the results from the five different types of models for the same forecast length. For forecast results with CC values more than 0.5 (we arbitrarily select this value as the “meaningful” forecast result), the differences in CC between different models are almost not observable. It is noted that there is a slight rise in CC for the buoy 45005 when the forecast length increases from 48h to 72h. This is simply due to the fluctuation of autocorrelation of the data when the time lag is large. The lead-lag correlation coefficient will fluctuate between 0 ~ 0.2 when the time lag of the autocorrelation varies from 40~240h (Figure S2 in Supplementary Material). Different models has also different CCs in 72-h forecast at buoy 45005 (which is probably the most clear difference between different subplots). However, such a low correlation is statistically insignificant and not useful for forecast practice, and this difference only corresponds to an RMSE difference of ~0.01m.

Furthermore, we examined two arbitrarily selected examples of SWH time series to analyze whether there is a significant difference in the prediction mechanism between the traditional AR model and deep learning models. Figure 4 shows the comparison of the model-predicted and buoy-observed SWH time series at buoy 44007 and 46041 in May 2017. The results for different forecast lengths (6h, 12h, 24h, 72h) from both the AR model and ANN model are shown (the input lengths for both models are 24h). The reason for selecting the ANN model here is that all deep neural networks are essentially subsets of ANN. When the performance of these models is nearly identical, the neural networks will also work in almost the same way. The curves of the AR model and ANN model agree very well: when the forecast duration is short (less than 12h), there is an obvious lag relationship between the predicted results of both models and the observed results, and the lag time is the same as the forecast length. As the forecast length gradually increases, the time lag of the predicted results also gradually lengthens, and the predicted results gradually converge toward the mean value of the time series. When the forecast length at buoy 44007 reaches 72h, both models have completely lost their prediction ability, and can only use the mean value

210 as the predicted result to minimize the RMSE. In other words, all the models show the same property at
 211 all buoy locations: With the extension of forecast length, the predicted results of all models gradually
 212 transform from nearly inertial prediction to the mean value of the time series.

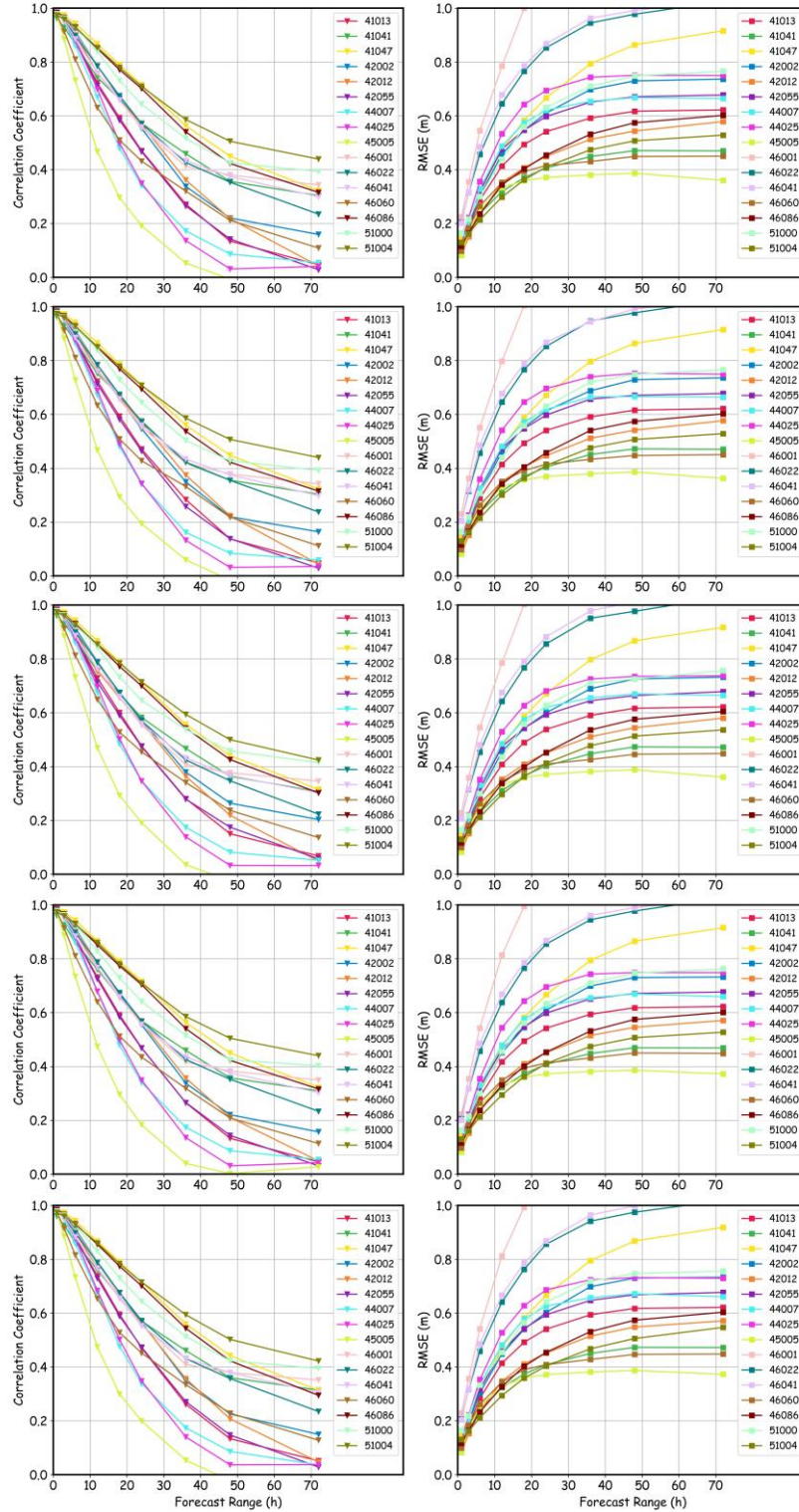


Figure 3. The variation of error metrics (left: CC, right: RMSE) with forecast length for five models (from the top to the bottom: AR, XGB, ANN, LSTM, WaveNet) at 16 buoy locations.

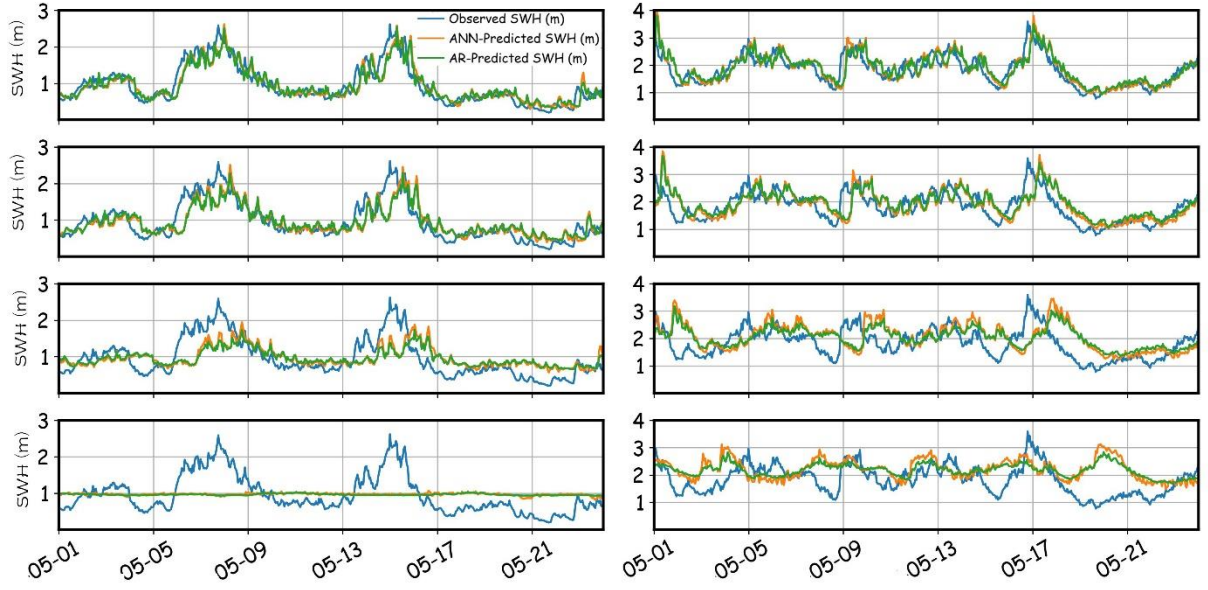


Figure 4. Comparison of the SWH time series predicted by the AR and ANN with the observed SWH at buoy 44007 (left) and 46041 (right) in May 2017. The input length is 24h, and the four rows correspond to the forecast lengths of 6h, 12h, 24h, and 72h, respectively.

4 Discussions

As previously noted, the essence of univariate time series prediction is to explore the linear or nonlinear autocorrelation within the sequence. No time series prediction tool can handle completely random time series with no autocorrelation in the signal. Given the input length j and forecast length i , the form of the time series prediction model can be:

$$X_{n+i} = f(X_n, X_{n-1}, X_{n-2}, \dots, X_{n-j}) + \delta \quad (3)$$

where δ is the random error of the model. The function f can be linear or strongly nonlinear. As a linear model, the AR model can only handle linear components within the autocorrelation:

$$X_{n+i} = a + a_0X_n + a_1X_{n-1} + a_2X_{n-2} + \dots + a_jX_{n-j} + \delta \quad (4)$$

This form is limited in handling nonlinear components. In contrast, various AI models can learn nonlinear forms of autocorrelation components from data due to their powerful non-linear fitting capabilities. However, if the non-linear autocorrelation components within the sequence are weak compared to the linear autocorrelation and random components, AI models do not have significant advantages over AR models.

It is noted that the non-linear component of autocorrelation within the SWH sequence and the nonlinearity of the SWH sequence itself are different concepts. The nonlinearity of the SWH sequence

refers to its inability to be generated by a linear function, necessitating consideration of the random error term δ in Eq (4). The non-linear autocorrelation component refers to the nonlinear component contained within f of Eq (3). In other words, if we further decompose Eq (3) into:

$$X_{n+i} = a + a_0X_n + a_1X_{n-1} + a_2X_{n-2} + \dots + a_jX_{n-j} + g(X_n, X_{n-1}, X_{n-2}, \dots, X_{n-j}) + \delta \quad (5)$$

Then, the g in Eq (5) represents the nonlinear component of autocorrelation.

From the result, it can be found that the nonlinear components in the autocorrelation of the SWH time series are not significant. The AI-based models used in this study can only learn the linear autocorrelation of the sequence from the data, so its predicted results are almost the same as those of the AR model. For the specific problem of SWH time series short-term prediction, considering that the AR model has less complexity and better interpretability, we think the AR model may be even superior to AI models to some extent.

Some previous studies (e.g., Zhou et al. 2021, Huang and Dong 2021, Çelik 2022, Ding et al. 2023, Song et al. 2023, Lv et al. 2023, Chen et al. 2023, Altunkaynak et al. 2023, to name but a few) decomposed the SWH time series using signal decomposition methods, such as Empirical Mode Decomposition (EMD), before predicting the time series. They trained and predicted each Intrinsic Mode Function (IMF) or signal components separately to achieve better results. However, it is essential to highlight that THIS METHOD IS INCORRECT and IMPRACTICAL. These studies often decomposed the whole time series in both training and test sets (either together or separately), and then used the trained models to forecast the corresponding components in the test set. However, signal decomposition can only be performed on known time series. As the test set consists of unknown future data, decomposing it using EMD or other signal decomposition techniques in advance is unreasonable. In real-world applications, future SWH values cannot be known in advance (otherwise, prediction would be unnecessary), making it impossible to pre-decompose the "test set". Simply speaking, the signal decomposition should never be applied to the whole test set because the test set should be regarded as known. Since signal decomposition cannot be directly applied to the test set, one must decompose the time series up to the current moment before each prediction and then compare the result with the corresponding value in the test set. However, no signal decomposition-based studies adopted this (correct) iteratively procedure to achieve satisfactory results, to the best of our knowledge. This mistake of decomposing test set is common not only in SWH

time series prediction but also in many works of other time series prediction (e.g., Wang et al., 2021, Ha et al., 2021, Yan et al. 2020, Han et al. 2023, also to name but a few).

Considering these “mainstream” AI algorithms do not significantly outperform AR models in predicting SWH time series, this study recommends that for future potential applications of new AI models to this problem, the results of an AR model with the same input and prediction lengths should serve as a benchmark. Likewise, for other univariate time series prediction problems, this study also recommends using the results of the simplest AR model with the same input and output as a benchmark for comparison.

5 Summary

Deep learning (or AI) is a powerful tool for statistically modelling of many phenomena including ocean waves, as demonstrated by numerous previous studies (e.g., Liu et al. 2024, Wang et al. 2023, Song and Jiang 2023). However, it does not necessarily outperform the simplest AR model for simple regression problems such as univariate SWH time series prediction. Using the in-situ observations from 16 buoys, this study compared the performance of five models - AR, XGB, ANN, LSTM, and WaveNet – on the problem of SWH time series short-term prediction. Surprisingly, the results show that for a given input and prediction lengths, there is no significant difference in the accuracy among these models, which is a finding not previously recognized. This is because the nonlinear components in the autocorrelation of SWH time series are much weaker than the linear components and random components, leading to the result that all AI-based models can only “learn” the linear autocorrelation components in the time series in this specific issue. Considering the complexity, stability, and interpretability of modeling, AR seems to be a better selection with respect to SWH time series prediction. Since almost all previous works of SWH time series forecast based on AI did not realize this issue, this paper can be regarded as a comment on them. As a comment, we also pointed out that applying signal decomposition to the time series before prediction, which is common in previous studies, is also wrong because they decompose the test set.

Despite the negative results, this study suggests that AR can be regarded as a benchmark for future SWH (and potentially other variables) time series prediction models. The results also implicate that time

series with linear components in autocorrelation much more significant than non-linear components may not common in nature. When predicting time series of various ocean environmental parameters (including SWH), the prediction results of the AR model should be regarded as a baseline: An AI-based time series prediction model can be deemed effective only if its performance significantly outperforms the AR model. Although our analysis only focuses on univariate time series prediction, we believe this benchmark can be also employed for simple multivariate-input time series prediction when the AR model is easily applicable.

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The NDBC buoy data are available from the website of the National Centers for Environmental Information (<https://www.ncei.noaa.gov/data/oceans/ndbc/cmanwx/>). A ready-to-use version of the data and the corresponding Python code is provided in the Supplemental Materiel. This work was jointly supported by the National Key Research and Development Program of China (2023YFC3008203), the National Natural Science Foundation of China (42376172), and the Guangdong Basic and Applied Basic Research Foundation (2022A1515240069).

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